

UCI: Division of Continuing Education
Accelerated Certificate Program (ACP)
Data Science
Introduction to Python Programming
I&C SCI_X426.58:
Homework#8

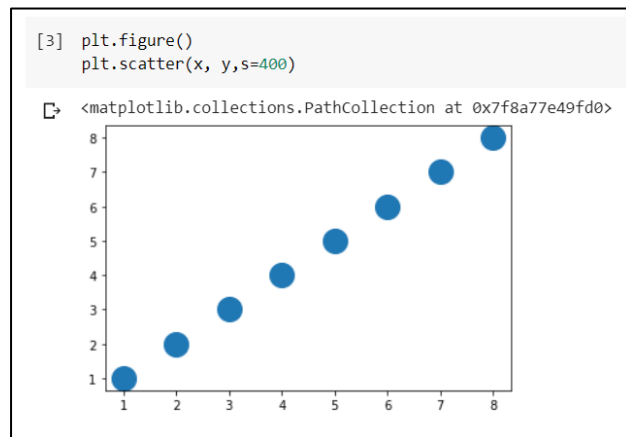
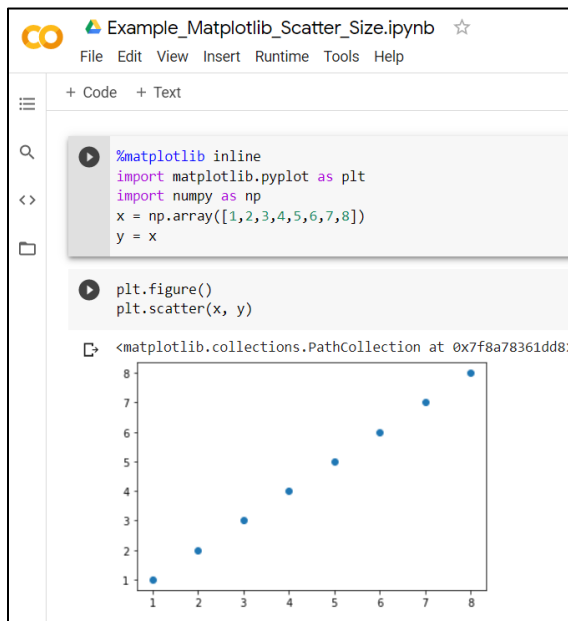
Date Given: April 14, 2023

Due Date: April 17, 2023

Creating a 3-dimensional plot using Python (Matplotlib)

A “Scatter bubble chart” depicts three-dimensional data through the position (x and y coordinates) and size of the marker. The `Matplotlib.pyplot`’s `scatter` method can produce bubble charts by passing the marker size to its ‘s’ attribute in (*points*)² such that the area of the marker is proportional to the magnitude of the third dimension.

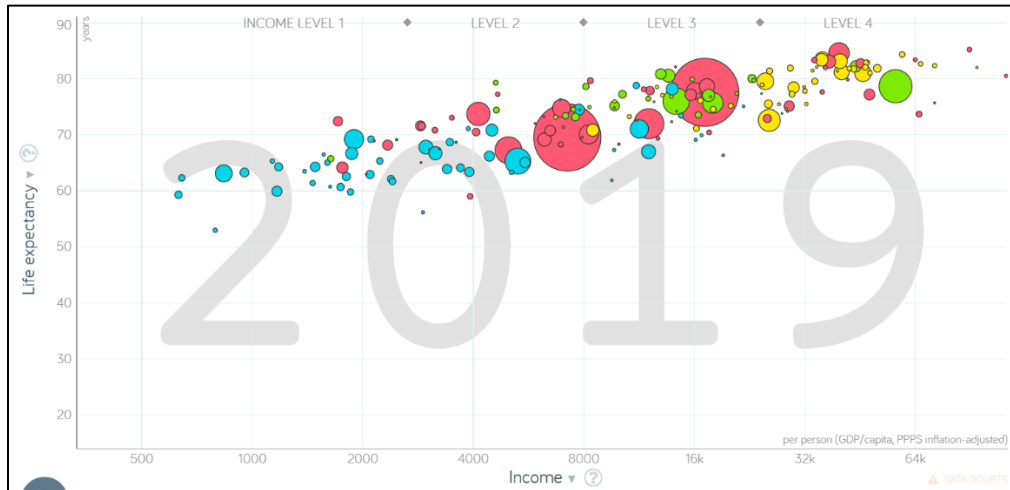
Analyze the following 2 plots. The left side scatter plot is created without the ‘size’ attribute. The right-side scatter plot is created with the ‘size’ attribute equal to 400.



The <https://www.gapminder.org/> website displays many 3-dimensional plots using the “Scatter bubble chart” technique. The following plot is an example of a “Scatter bubble chart” taken from the gapminder.org website.

[https://www.gapminder.org/tools/#\\$chart-type=bubbles](https://www.gapminder.org/tools/#$chart-type=bubbles)

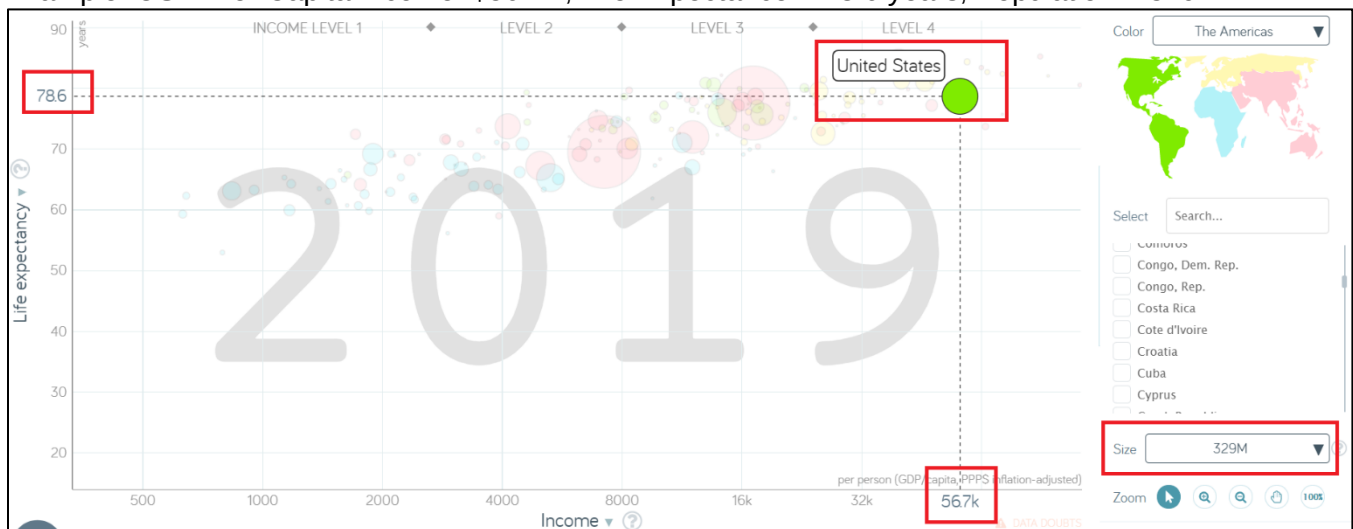
The following chart displays the information about all the countries using the data collected in 2019. The x-axis represents ‘per-capita income’, and the y-axis represents ‘life-expectancy’. The size of the bubble represents the country’s population.



This chart is also interactive. When a mouse hovers on any bubble it displays the following information about that country

- Per Capita Income
- Life Expectancy
- Population

Example: USA: Per Capita Income: \$56.7K, Life Expectance = 78.6 years, Population = 329M



Problem#1

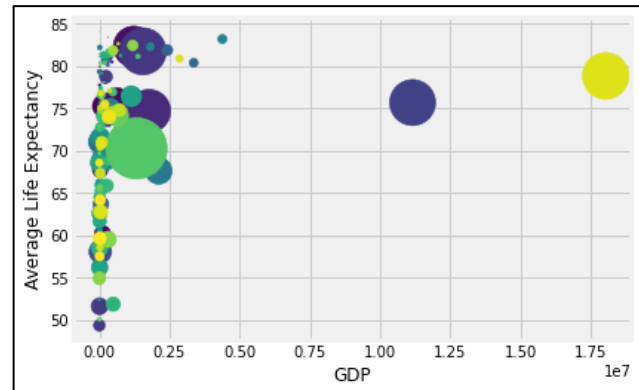
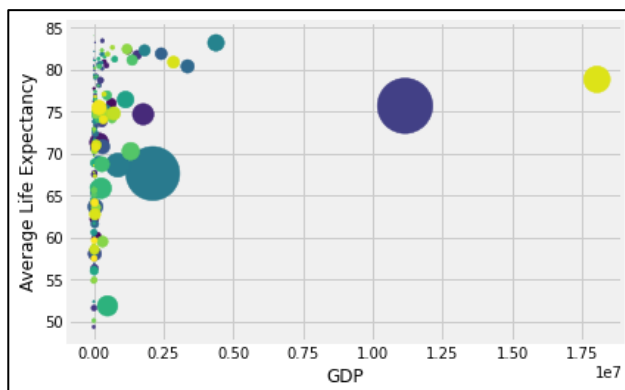
Bubble Chart for 3-dimensional data

Read the dataset 'country_profile_variables.csv' which contains data about the 212 countries. This dataset was downloaded from UN's (United Nations) website.

	A	B	C	D	E	F	G
1		Country	Surface area (km2)	Population 1,000 (2017)	GDP (M US\$)	Life Exp Female	Life Exp Male
2	1	Afghanistan	652864	35530	20270	63.5	61
3	2	Albania	28748	2930	11541	79.9	75.6
4	3	Algeria	2381741	41318	164779	76.5	74.1
5	4	American Samoa	199	56	-99	77.8	71.1
6	5	Angola	1246700	29784	117955	63	57.4
7	6	Antigua and Barbuda	442	102	1356	78.2	73.3
8	7	Argentina	2780400	44271	632343	79.8	72.2
9	8	Armenia	29743	2930	10529	77	70.6
10	9	Aruba	180	105	2702	77.8	72.9
11	10	Australia	7692060	24451	1230859	84.4	80.2
12	11	Austria	83871	8736	376967	83.5	78.4
13	12	Azerbaijan	86600	9828	53049	74.6	68.6
14	13	Bahamas	13940	395	8854	78.1	72

- 1.1 Compute the difference between the life expectancy of males and females for all the countries. Display the top-10 countries where is the difference of life expectancy between males and females is the highest. Display the difference of life expectancy between males and females in the 'United State of America'.
- 1.2 Compute the "average life expectancy" of males and females for all the countries. We will use this variable in the next 2 problems (1.3 + 1.4).
- 1.3 Create 2 scatter bubble charts between GDP and "average life expectancy" for all the countries in the dataset.
 - In the first chart the size of the bubble should be proportional to the population of the country.
 - In the second chart the size of the bubble should be proportional to the surface area of the country.

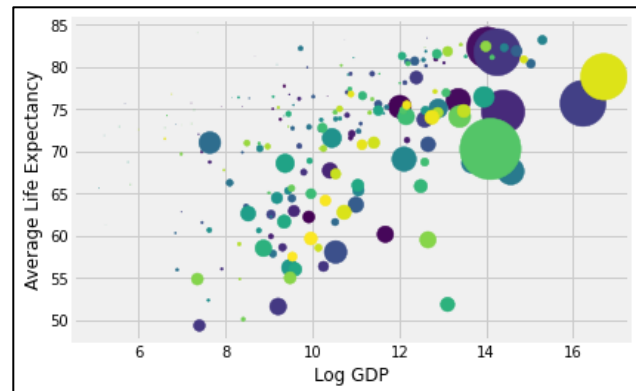
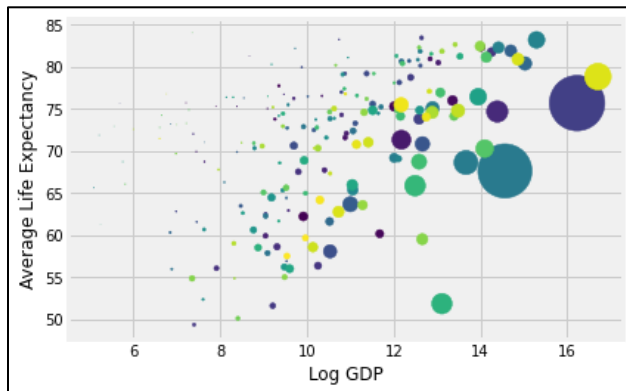
Your plots will look as follows.



1.4 The plots created in the previous problem (1.3) have bubbles squeezed on the left side of the graph. Create 2 more scatter bubble charts between 'log (natural log) value of GDP' and "average life expectancy" for all the countries in the dataset. This will allow the bubbles to spread out all over the graph.

- In the first chart the size of the bubble should be proportional to the population of the country.
- In the second chart the size of the bubble should be proportional to the surface area of the country.

Your plots will look as follows.



```
import pandas as pd
import numpy as np
```

```
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
%matplotlib inline
```

```
from google.colab import files
f = files.upload()
```

country_pro...riables.csv

- **country_profile_variables.csv**(text/csv) - 9104 bytes, last modified: 10/29/2019 - 100% done
Saving country_profile_variables.csv to country_profile_variables.csv

```
# Step# 1
#
```

```
country = pd.read_csv('country_profile_variables.csv')
print(country.head(5))
print(country.tail(5))
print(country.shape)

print(type(country))
```

```

Country Surface area (km2) Population 1,000 (2017) \
0 1 Afghanistan 652864 35530
1 2 Albania 28748 2930
2 3 Algeria 2381741 41318
3 4 American Samoa 199 56
4 5 Angola 1246700 29784

GDP (M US$) Life Exp Female Life Exp Male
0 20270 63.5 61.0
1 11541 79.9 75.6
2 164779 76.5 74.1
3 -99 77.8 71.1
4 117955 63.0 57.4

Unnamed: 0 Country Surface area (km2) \
206 207 Wallis and Futuna Islands 142
207 208 Western Sahara 266000
208 209 Yemen 527968
209 210 Zambia 752612
210 211 Zimbabwe 390757

Population 1,000 (2017) GDP (M US$) Life Exp Female Life Exp Male
206 12 -99 78.7 72.8
207 553 -99 70.3 66.9
208 28250 29688 65.6 62.8
209 17094 21255 61.9 57.5
210 16530 13893 59.0 56.1
(211, 7)
<class 'pandas.core.frame.DataFrame'>
```

```
lifeExpFemale = country['Life Exp Female']
lifeExpMale = country['Life Exp Male']
```

```
diffLE_MaleFemale = lifeExpFemale - lifeExpMale
averageLifeExp = (lifeExpFemale + lifeExpMale)/2
print(averageLifeExp)
```

```

0 62.25
1 77.75
2 75.30
3 74.45
4 60.20
...
206 75.75
207 68.60
208 64.20
209 59.70
210 57.55
Length: 211, dtype: float64
```

```
country['averageLifeExp'] = averageLifeExp
country['Difference'] = diffLE_MaleFemale
```

```
print(country.head(5))
```

Unnamed: 0	Country	Surface area (km2)	Population 1,000 (2017)	\
0	1 Afghanistan	652864	35530	
1	2 Albania	28748	2930	
2	3 Algeria	2381741	41318	
3	4 American Samoa	199	56	
4	5 Angola	1246700	29784	

	GDP (M US\$)	Life Exp Female	Life Exp Male	averageLifeExp	Difference
0	20270	63.5	61.0	62.25	2.5
1	11541	79.9	75.6	77.75	4.3
2	164779	76.5	74.1	75.30	2.4
3	-99	77.8	71.1	74.45	6.7
4	117955	63.0	57.4	60.20	5.6

```
sortedTable = country.sort_values(['Difference'],ascending=False)
print(sortedTable.head(5))
print(sortedTable.shape)
sortedTable.loc[sortedTable['Country'] == 'United States of America']
```

Unnamed: 0	Country	Surface area (km2)	\
182	183 Syrian Arab Republic	185180	
154	155 Russian Federation	17098246	
16	17 Belarus	207600	
107	108 Lithuania	65286	
102	103 Latvia	64573	

	Population 1,000 (2017)	GDP (M US\$)	Life Exp Female	Life Exp Male	\
182	18270	28393	76.3	64.4	
154	143990	1326016	75.9	64.7	
16	9468	54609	77.7	66.5	
107	2890	41402	79.3	68.5	
102	1950	27004	78.7	68.8	

	averageLifeExp	Difference
182	70.35	11.9
154	70.30	11.2
16	72.10	11.2
107	73.90	10.8
102	73.75	9.9

(211, 9)

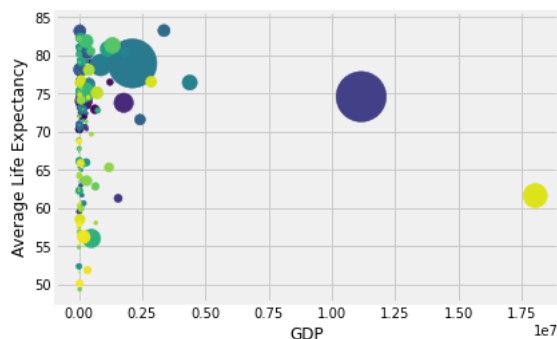
Unnamed: 0	Country	Surface area (km2)	Population 1,000 (2017)	GDP (M US\$)	Life Exp Female	Life Exp Male	averageLifeExp	Differer
	United							

```
countryName = country['Country']
GDP = country['GDP (M US$)']
averageLifeExp = sortedTable['averageLifeExp']
```

```
marker_size_P = country['Population 1,000 (2017)']/1000 # Proportional to the Population
marker_size_SA = country['Surface area (km2)']/10000 # Proportional to the Surface Area
```

```
fig, ax = plt.subplots()
```

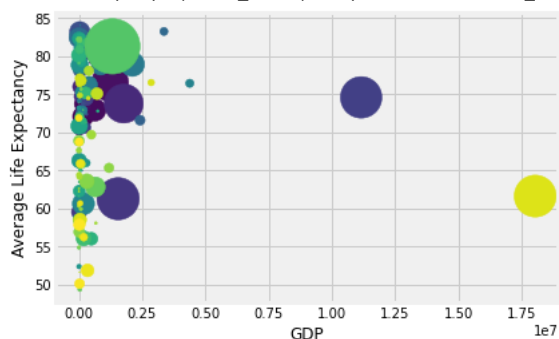
```
ax.scatter(GDP, averageLifeExp, s=marker_size_P, c = range(len(countryName)))
ax.set_xlabel("GDP")
ax.set_ylabel("Average Life Expectancy")
plt.show()
```



```
fig, ax = plt.subplots()
```

```
ax.scatter(GDP, averageLifeExp, s=marker_size_SA, c = range(len(countryName)))
ax.set_xlabel("GDP")
ax.set_ylabel("Average Life Expectancy")
plt.show()
```

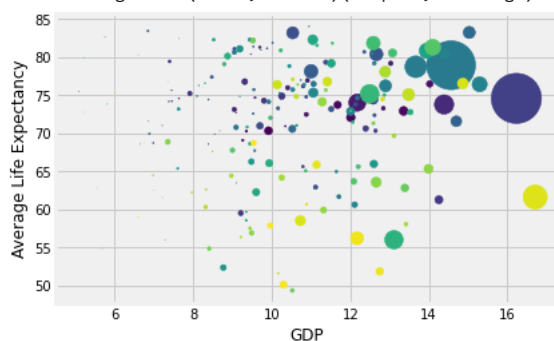
```
/usr/local/lib/python3.9/dist-packages/matplotlib/collections.py:963: RuntimeWarning: invalid
scale = np.sqrt(self._sizes) * dpi / 72.0 * self._factor
```



```
fig, ax = plt.subplots()
```

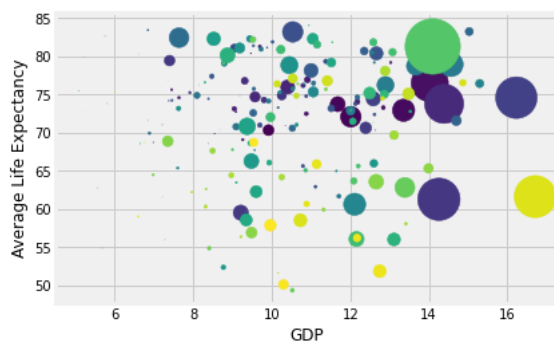
```
ax.scatter(np.log(GDP), averageLifeExp, s=marker_size_P, c = range(len(countryName)))
ax.set_xlabel("GDP")
ax.set_ylabel("Average Life Expectancy")
plt.show()
```

```
/usr/local/lib/python3.9/dist-packages/pandas/core/arraylike.py:397: RuntimeWarning: invalid
result = getattr(ufunc, method)(*inputs, **kwargs)
```



```
fig, ax = plt.subplots()
```

```
ax.scatter(np.log(GDP), averageLifeExp, s=marker_size_SA, c = range(len(countryName)))
ax.set_xlabel("GDP")
ax.set_ylabel("Average Life Expectancy")
plt.show()
```



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